



B.TECH — INFORMATION TECHNOLOGY | COLLEGE PROJECT

Community-Detection Platform

A Graph-Based Analytical System for Uncovering Hidden Communities in Social, Communication & Collaboration Networks

Presented by:

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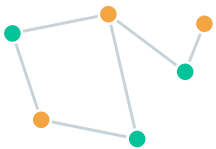
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Why Community Detection Matters

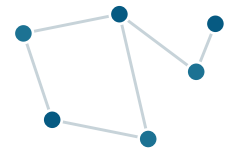
Introduction

Community Detection Platform is a graph-based system designed to identify groups of closely connected users within a network. In social, communication, and collaboration networks, users interact in different ways, creating complex relationship patterns. By representing users as nodes and their interactions as edges, the system analyzes these relationships and identifies meaningful communities.



Problem Statement

- Modern networks contain thousands to millions of interactions — growing continuously.
- Manually identifying meaningful groups or influencers is impractical at this scale.
- Raw interaction logs don't reveal structure on their own, and existing tools rarely visualize it well.
- Need: an automated, scalable platform to detect and visualize communities directly from data.

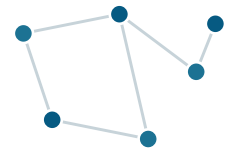
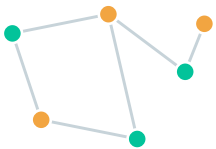


ABSTRACT

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The Community Detection Platform analyzes user interaction data to discover naturally formed groups within a network. The system constructs a weighted graph where nodes represent users and edges represent relationships, with weights indicating the strength of interaction. A community detection algorithm is then applied to identify groups whose members are more strongly connected than to the rest of the network. The detected communities are visualized to provide insights into network structure, user relationships, and interaction patterns. The proposed system can be applied to social network analysis, recommendation systems, collaboration networks, and customer segmentation.



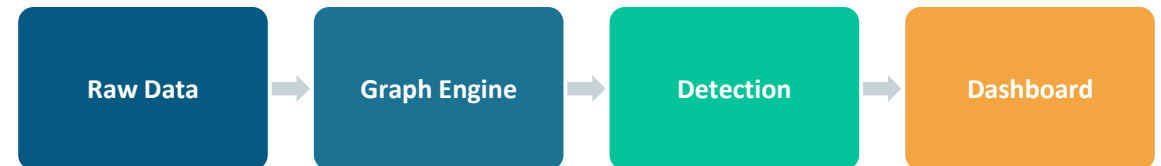
What We Aim to Build

Objectives

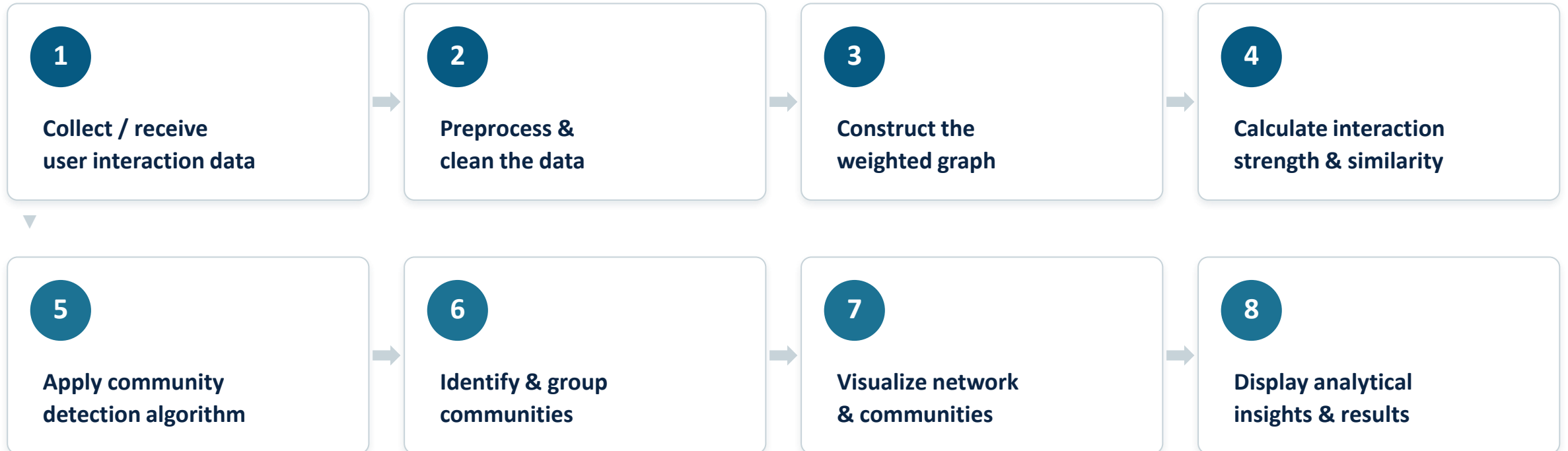
- 1 Construct a weighted interaction graph from raw user activity data.
- 2 Quantify interaction strength via frequency, communication pattern & content similarity.
- 3 Apply an efficient community detection algorithm to partition the network.
- 4 Visualize the network and detected communities interactively.
- 5 Generate insights on influential users, group sizes & density.

Proposed Solution

- End-to-end platform: ingest interaction data, output an interactive community map.
- Core engine built on a weighted graph model (NetworkX) with a modular detection pipeline.
- Web dashboard (Flask / Streamlit) for uploading data, running analysis & exploring results.
- Scales from small course-project datasets to larger real-world samples.

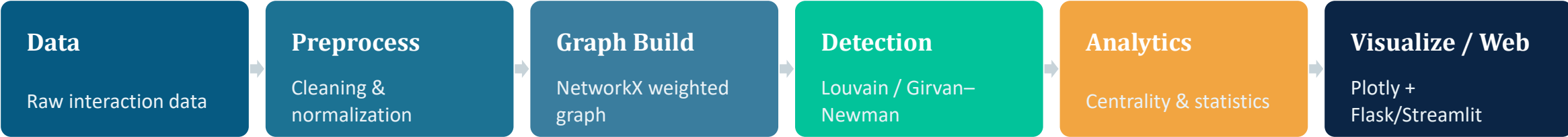


Step-by-Step Project Workflow



Each stage feeds the next — from raw interaction records to a fully visualized, analyzable community map.

How the Pieces Fit Together



Technology	Purpose
Python 3.x	Core language for data processing & algorithms
NetworkX	Graph construction & built-in community detection
Pandas / Scikit-learn	Data cleaning & content-similarity computation
Flask / Streamlit	Interactive web dashboard
MongoDB / MySQL	Stores raw data & detection results
Matplotlib / Plotly	Graph & analytics visualization

Minimum Hardware

- i5 / Ryzen 5 processor
- 8 GB RAM (16 GB recommended)
- 256 GB SSD, 10 GB free
- Windows 10/11, macOS, or Ubuntu
- GPU optional; internet required for setup

LITERATURE SURVEY

Author / Research	Contribution	Relevance to Our Project
Newman	Introduced modularity-based evaluation of community structures	Helps evaluate the quality of detected communities
Girvan & Newman	Proposed edge betweenness-based community detection	Provides a foundational approach to network partitioning
Blondel et al.	Introduced the Louvain Method for modularity optimization	Suitable for efficient community detection
Weighted Network Research	Uses interaction strength as edge weights	Supports stronger and more realistic relationship analysis
Network Visualization Research	Focuses on visual representation of graphs and communities	Helps display and interpret detected groups

Thank You